

Mid-term test

CSC354 · Machine Learning

TOTAL MARKS 100	DURATION 30 min	DIFFICULTY MEDIUM	THRESHOLD 75%	QUESTIONS 10
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STUDENT NAME	STUDENT ID	SECTION	DATE
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INSTRUCTIONS

- Read all questions carefully before answering.
- All questions are compulsory unless stated otherwise.
- Total time allowed: 30 minutes.
- No electronic devices allowed during the quiz.
- Write clearly and legibly in the space provided.

Questions

Assessment Questions

- 1 In supervised learning for classification problems in machine learning, which algorithm is commonly used to predict the category of a given input?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Principal Component Analysis
- ☐ D K-Means Clustering

Student Answer: &

- 2 Which supervised learning technique is most suitable for solving a binary classification problem?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Principal Component Analysis (PCA)
- ☐ D K-Means Clustering

Student Answer: &

- 3 Which of the following is a common supervised learning technique used for solving classification problems in machine learning?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Principal Component Analysis
- ☐ D K-Means Clustering

Student Answer: &

4 In supervised learning for classification problems in machine learning, which algorithm is commonly used to predict the class of given input data?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Principal Component Analysis
- ☐ D K-Means Clustering

Student Answer: &

5 Which supervised learning technique is most commonly used for solving binary classification problems?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Decision Trees
- ☐ D K-Means Clustering

Student Answer: &

6 In supervised learning for classification problems, which algorithm is commonly used to predict categorical outcomes?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Decision Trees
- ☐ D Principal Component Analysis

Student Answer: &

7 In supervised learning for classification problems in machine learning, which algorithm is commonly used to predict categorical outcomes based on input features?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Decision Trees
- ☐ D K-Means Clustering

Student Answer: &

8 In supervised learning for classification problems in machine learning, which algorithm is commonly used to predict the class of a given input?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Principal Component Analysis (PCA)
- ☐ D K-Means Clustering

Student Answer: &

9

Which supervised learning technique is most suitable for solving a binary classification problem with imbalanced classes?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression
- ☐ C Decision Trees without any modifications
- ☐ D Naive Bayes Classifier

Student Answer: &

10

Which supervised learning technique is most suitable for solving a binary classification problem where the classes are imbalanced?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Logistic Regression with class weights adjustment
- ☐ C Naive Bayes Classifier without any modification
- ☐ D K-Nearest Neighbors (k=1)

Student Answer: &